

How predictable are rare precipitation extremes? A calibration-aware assessment for London and Paris

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Abstract

How predictable are rare precipitation extremes, and how should that predictability be measured? We address both questions using thirty years (1989–2018) of a standardised precipitation-extreme index for London (maritime) and Paris (more continental), with strictly antecedent ERA5 drivers. We argue rare-event forecasts must be judged by *calibration-aware proper scores* against climatology, and assemble a protocol (Brier decomposition, threshold-weighted CRPS skill, bootstrap intervals, multi-split checks) guarding against the single-seed and quantile-selection pitfalls that inflate apparent skill. Across a bake-off from logistic regression to a Gaussian-process extreme-value hurdle, we find, consistently in both cities, that the Gaussian-process hurdle is the most skilful full-distribution forecaster on the tail, while simple calibrated models match it on occurrence, and the choice of response distribution matters as much as model class. Occurrence is weakly but genuinely predictable from antecedent convective instability and circulation, and the magnitude of the largest events is weakly but robustly predictable from convective instability, with a CRPS skill over climatology excluding zero across four successive splits per city. The effects are small, statistically robust, and physically plausible, with replication across two cities offering modest evidence against a location-specific artefact.

Keywords: extreme-value theory; proper scoring rules; forecast verification; conformal prediction; Gaussian processes; convective instability.

1 Introduction

Extreme precipitation is among the most damaging natural hazards facing temperate cities. Short, intense rainfall episodes trigger surface-water and river flooding, disrupt transport, overwhelm drainage, and impose large and unevenly distributed economic and social costs. In cities such as London and Paris such events occur on only a small fraction of days, yet a single episode can cause disproportionate harm, and both observational records and climate projections indicate that the frequency and intensity of heavy rainfall are increasing (IPCC, 2021). Skilful and, above all, *trustworthy* early warning of these events would let emergency managers, infrastructure operators, and insurers act pre-emptively. This motivates probabilistic forecasts of extreme precipitation at horizons of days to weeks ahead.

The statistical obstacles are severe. Extreme days are rare, typically a few percent of the record, so a model that simply predicts “no extreme” is accurate yet useless, and conventional accuracy is the wrong yardstick. The daily precipitation index is zero-inflated, strongly right-skewed, and heavy-tailed, violating the Gaussian assumptions underlying standard regression. The observational record is short relative to the rarity of the events, leaving few positive examples from which to learn, and the day-to-day persistence of extremes is limited, so intrinsic predictability is modest. Together these features cap raw forecast *skill*, a ceiling any honest treatment must confront rather than obscure.

We argue that, precisely because raw predictability is limited, the decisive quality of a forecast in this setting is not discrimination but *calibration*: whether the issued probabilities can be trusted. A forecast that assigns a 5% probability to an extreme is useful for decision-making only if extremes do occur on about 5% of such days; an overconfident forecast that systematically overstates rare-event probabilities causes false alarms and wasted resources, while an underconfident one breeds complacency. The appropriate goal is therefore to *maximise sharpness subject to calibration* (Gneiting et al., 2007): to be as informative as the data allow while remaining honest. Calibration matters most exactly where it is hardest to achieve — in the tail — and evaluating it there is itself subtle, as the *forecaster’s dilemma* makes clear (Lerch et al., 2017).

No existing methodology delivers calibrated, guaranteed tail probabilities for rare-event forecasting, because the three relevant tools each supply only part of the answer. *Extreme-value theory* (EVT) furnishes asymptotically justified tail models, but is most often used descriptively, for return levels or non-stationary tail regression, rather than for flexible, covariate-driven probabilistic forecasting, and it offers no finite-sample coverage guarantee (Coles, 2001; Davison and Smith, 1990). *Flexible machine learning*, including deep models and Gaussian processes, captures non-linear dependence on atmospheric predictors and yields predictive distributions, but those distributions are typically miscalibrated in the tail and come with no guarantee (Galib et al., 2022; Lakshminarayanan et al., 2017). *Conformal prediction* provides distribution-free, finite-sample coverage, but its standard form does not guarantee conditional or tail-specific coverage — the effective resolution of very high quantiles is limited by the size of the calibration set — and assumes exchangeability, which is generally violated in daily time series due to temporal dependence and non-stationarity (Gibbs and Candès, 2021; Pasche et al., 2025).

A further, orthogonal limitation cuts across these tools: many two-part (hurdle) forecasting approaches estimate occurrence and intensity in separate stages, so that inference about extremes relies primarily on the relatively small number of exceedances. This separation can limit the extent to which information contained in the abundant non-extreme data informs tail inference, particularly when common latent processes govern both occurrence and intensity.

Rather than advocate a single model, we therefore ask a more basic pair of questions — *how predictable are these extremes, and how should that predictability be measured?* — and answer them with a careful evaluation protocol and a bake-off of models spanning logistic regression, zero-inflated distributional regressions (log-normal, gamma, and the threshold-free extended generalised Pareto distribution), and a Gaussian-process extreme-value hurdle with conformal calibration.

Contributions.

1. A calibration-aware evaluation protocol for rare-event precipitation forecasting: Brier reliability/resolution decomposition, threshold-weighted continuous ranked probability score (CRPS), skill scores against climatology, bootstrap confidence intervals, and multi-split robustness – designed to resist the single-seed and quantile-selection artefacts that inflate apparent skill, and to demonstrate that conformal coverage is insufficient as a stand-alone measure of forecast quality.
2. A rigorous predictability assessment, replicated in two contrasting cities: occurrence of an extreme is weakly but genuinely predictable from antecedent state; and the *magnitude* of the largest events carries weak but robust predictability tied to antecedent convective instability, with a CRPS skill over climatology whose moving-block bootstrap interval excludes zero across four successive train/test splits in each city – and is stronger in the more convective Paris, consistent with a meteorological rather than a purely local origin (though two cities in one climate zone offer only modest evidence on geographic generality).
3. A methodological finding on what matters: across the bake-off, a Gaussian-process extreme-

value hurdle is the most skilful full-distribution forecaster on the tail in both cities – and, on occurrence, is matched by an inexpensive calibrated logistic regression and gradient-boosted trees – while the choice of response distribution matters as much as model class (a zero-inflated log-normal is catastrophically miscalibrated where a threshold-free extended generalised Pareto distribution is well behaved), and conformal coverage, though attained by construction, is insufficient on its own as a measure of forecast quality.

Empirically, we study the daily standardised precipitation index over 1989–2018 for two cities of contrasting climate: London (maritime, 417 extreme days at the 95th percentile) and Paris (more continental and convective, 607 extreme days). The 95th-percentile index is the primary target; the rarer 99th-percentile index (137 events for London) is too data-scarce for the held-out intensity analysis (its occurrence-only results are in the Supplementary Material). Predictors are strictly antecedent ERA5 atmospheric drivers (circulation, moisture, and convective-instability fields) and large-scale teleconnection indices. Because the index is itself an ERA5-derived precipitation product, we exclude all precipitation fields from the predictors and lag every driver, so the assessment is of genuine antecedent predictability rather than diagnosis.

The remainder of the paper is organised as follows. Section 2 positions our work within the literatures on extreme-value modelling, Gaussian processes, conformal prediction, and forecast evaluation. Section 3 describes the data and predictors. Section 4 sets out the models we compare, the conformal calibration, and the evaluation protocol. Section 5 presents the results, and Sections 6–7 discuss implications and conclude.

2 Related work

Our work sits at the intersection of four literatures; we review each in turn and then state precisely how this study differs.

Statistical modelling of precipitation extremes. Extreme-value theory provides the asymptotic foundation for tail modelling. The peaks-over-threshold approach models exceedances above a high threshold by the Generalised Pareto distribution (GPD) (Pickands, 1975; Davison and Smith, 1990), the framework we adopt for the intensity component; the block-maxima approach uses the Generalised Extreme Value (GEV) distribution (Coles, 2001). EVT is widely applied in hydrology and climatology (Katz et al., 2002), including non-stationary and spatial Bayesian formulations of precipitation extremes (Cooley et al., 2007). This body of work is, however, predominantly *descriptive* – estimating return levels or letting tail parameters vary smoothly with a few covariates – rather than producing flexible, covariate-rich *forecasts* at a fixed lead time, and it offers no finite-sample coverage guarantee. We retain the GPD tail but embed it in a flexible predictive model and calibrate it conformally.

Machine learning and deep learning for extremes. Data-driven methods increasingly target extremes: Galib et al. (2022) forecast block maxima with a deep network constrained by GEV theory, and neural networks have been used to estimate extreme-value parameters. More broadly, predictive uncertainty in deep models is quantified with deep ensembles (Lakshminarayanan et al., 2017) and Monte-Carlo dropout (Gal and Ghahramani, 2016). These methods capture non-linear dependence on predictors and return uncertainty estimates, but they are typically miscalibrated in the tail and lack distribution-free guarantees – the gap our conformal layer closes.

Gaussian processes: scalability and multiple outputs. Gaussian processes offer principled non-parametric regression with built-in uncertainty (Rasmussen and Williams, 2006), made scalable through sparse variational inference with inducing points (Titsias, 2009; Hensman et al.,

2013; Salimbeni and Deisenroth, 2017). Expressiveness can be increased by composing layers in deep Gaussian processes (Damianou and Lawrence, 2013) or by learning the kernel through a neural feature map in deep kernel learning (Wilson et al., 2016), though as we find below such added depth is not warranted on our low-signal, single-site problem. Correlated outputs are modelled by multi-output GPs and coregionalisation (Bonilla et al., 2008; Álvarez et al., 2012). A GP with a Gaussian likelihood cannot represent heavy-tailed, skewed, zero-inflated responses; we therefore pair the GP prior with an EVT likelihood and use coregionalisation not to share information across tasks but to couple the two halves of a hurdle model.

Two-part, hurdle, and selection models. Semicontinuous and zero-inflated outcomes are commonly modelled by two-part (hurdle) decompositions separating occurrence from intensity. Allowing the two parts to be *dependent* has a long history: correlated random-effects formulations (Olsen and Schafer, 2001), copula couplings, and the sample-selection models of econometrics (Heckman, 1979), which explicitly address the correlation between a selection (occurrence) equation and an outcome (intensity) equation. To our knowledge, this dependence has not been exploited within a GP–EVT *forecasting* model, where intensity is observed only on the rare extreme days, making it natural to borrow strength from the abundant occurrence data. We implement and test this approach and report (in the Supplementary Material) that, at the current data volume, the coupling is only weakly identified and has little empirical impact.

Conformal prediction. Conformal prediction yields distribution-free, finite-sample coverage for any underlying predictor (Vovk et al., 2005). Split-conformal methods make this practical for regression (Lei et al., 2018), and conformalised quantile regression adapts the intervals to heteroscedasticity (Romano et al., 2019). Exchangeability can be relaxed for covariate shift (Tibshirani et al., 2019) and, crucially for us, for time series and online distribution shift through adaptive conformal inference (Gibbs and Candès, 2021; Xu and Xie, 2021). Coverage at very high confidence levels, the extreme regime, is addressed by Pasche et al. (2025), who combine EVT with conformal prediction to build reliable intervals for high-impact events. That work, the closest to ours, constructs *regression intervals* from a black-box extreme-quantile estimator under (weighted) exchangeability; it uses neither a Gaussian-process nor a generative model, does not forecast exceedance *probabilities*, and does not model temporal dependence natively. Here, by contrast, tail-targeted and temporally-adaptive conformal calibration is applied to a generative model, within a broader predictability study.

Forecast evaluation of extremes. Probabilistic forecasts are assessed with proper scoring rules (Gneiting and Raftery, 2007) under the paradigm of maximising sharpness subject to calibration (Gneiting et al., 2007); the Brier score admits a reliability–resolution decomposition (Murphy, 1973). For extremes, threshold-weighted scoring rules focus evaluation on the tail while remaining proper (Gneiting and Ranjan, 2011), and naive restriction to extreme observations can paradoxically reward miscalibrated forecasts — the forecaster’s dilemma (Lerch et al., 2017). Our evaluation is built on these principles.

Positioning. Much of this literature proposes ever more expressive models for extremes. We take the complementary stance of *measuring* how much is predictable and which modelling choices matter, under an evaluation protocol built from proper scoring rules, skill against climatology, and bootstrap and multi-split robustness. In doing so we find, against the prevailing direction, that a compact Gaussian-process extreme-value hurdle is the most *skilful* full-distribution forecaster in both cities, that simple calibrated models are competitive, and that conformal coverage alone is an unreliable guide to forecast quality (Section 6). This emphasis on rigorous predictability assessment, rather than a new architecture, is the gap we fill.

3 Data

3.1 The precipitation-extreme index

Our response variable is a standardised daily precipitation-extreme index, computed for two European cities, London and Paris, derived from the Copernicus Climate Data Store product *Extreme precipitation risk indicators for Europe and European cities (1950–2019)*. The underlying field is the ERA5 reanalysis dynamically downscaled to a $2\text{ km} \times 2\text{ km}$ grid (ERA5-2km) with urban parameterisation, which resolves London as 53 and Paris as 157 grid cells. For each cell and day the indicator is the standardised exceedance

$$\mathcal{I} = \frac{P}{P_q}, \quad (1)$$

where P is daily precipitation and P_q is the fixed q th percentile of wet days ($P \geq 1\text{ mm}$) over the 1950–2019 reference period. The index is dimensionless, and $\mathcal{I} > 1$ marks an extreme: precipitation exceeding the historical q th-percentile level. To obtain a single city-level series we take the daily maximum of $\mathcal{I}$ over the city’s cells, so an extreme day is one on which *any* part of the city exceeds its threshold. We analyse the thirty-year record 1989–2018 at the $q = 95$ (primary) and $q = 99$ thresholds. The two cities are chosen for their contrasting climates: London’s maritime regime versus Paris’s more continental, more convective one – a deliberate test of whether conclusions generalise.

3.2 Exploratory characteristics

The series comprises 10,957 daily observations. Its defining features – which directly motivate the modelling choices of Section 4 – are summarised in Table 1 and Figure 1, and are shared by both cities. First, extremes are *rare*: 3.8% of London days and 5.5% of Paris days exceed the 95th-percentile threshold (and only 1.3% at the 99th in London). Second, the index is *zero-inflated and heavy-tailed*: about a quarter of days are exactly zero, the median is below 0.02, yet the maximum exceeds 6 at the 95th-percentile standardisation – a strong right skew incompatible with a Gaussian response. Third, occurrence is *seasonal*: the monthly extreme rate peaks in summer (July–August, reaching 6% at the 95th percentile and 3% at the 99th) and is lowest in late winter and early spring, consistent with convective summer rainfall. Fourth, the extreme indicator exhibits only *limited temporal autocorrelation*, so the persistence available to an autoregressive forecaster is modest – one reason that drivers beyond the index’s own history (Section 3.3) are essential.

3.3 Antecedent predictors

All predictors are *strictly antecedent*: for a horizon- h forecast of day $t + h$ they are functions only of information available up to day t . We use three groups.

1. **Index history.** Lagged values of the index (up to 14 days), rolling means and maxima over 7/14/30-day windows, counts of recent exceedances, and harmonic encodings of day-of-year and month to capture seasonality.
2. **ERA5 atmospheric drivers.** Antecedent fields from the ERA5 reanalysis over a synoptic-scale window around each city: mean sea-level pressure and its gradients, geopotential height (500/850 hPa), specific and total-column water vapour, integrated water-vapour transport, near-surface temperature and winds, and convective-instability diagnostics – convective available potential energy (CAPE) and convective inhibition (CIN), the K-index, and the Total-Totals index. Fields are aggregated to daily statistics (means, and maxima for the convective indices) and reduced to regional summaries.

3. **Teleconnection indices.** Lagged North Atlantic Oscillation, Arctic Oscillation, and an El Niño–Southern Oscillation index, capturing large-scale circulation states relevant at longer horizons.

In total $d = 49$ predictors enter the full model: 20 index-history features (lags, rolling summaries, and seasonality harmonics), 26 ERA5 circulation, moisture, and convective-instability fields, and 3 teleconnection indices. In the analysis the drivers are introduced *cumulatively* – index history; + teleconnections; + ERA5 circulation and moisture; + convective instability – so that each increment isolates a class of predictors. All features are standardised using training-set statistics only; the complete list and the retrieval recipe are in Appendix A.

3.4 Threshold, training, and testing

We take the 95th-percentile index as the primary target: its 417 extreme days (287 train, 130 test) admit the full analysis, including the held-out intensity test, which requires extreme days in both training and test across several splits. The rarer 99th-percentile index has only 137 extreme days in thirty years (101 train, 36 test) – too few for the intensity analysis – so its occurrence-only results are given in Supplementary Section S1 (Table S1). Unless stated, results are for the 95th percentile.

Because the data are a time series, all splits respect chronology. The primary split trains on 1989–2008 (7,305 days) and holds out 2009–2018 (3,652 days); preprocessing statistics and conformal calibration sets are derived from the training period only. To verify that results are not an artefact of a single test window, the intensity analysis is additionally repeated over four forward-only, expanding-window splits (test periods beginning 2003, 2006, 2009, and 2012), and all stochastic results are averaged over random seeds.

4 Methodology

Section 4.1 fixes notation and the predictability target. Section 4.2 describes the ladder of models we compare, from climatology and logistic regression through zero-inflated distributional regressions to a Gaussian-process extreme-value hurdle. Section 4.3 describes the conformal calibration we assess (and why its coverage guarantee is, on its own, uninformative). Section 4.4 sets out the evaluation protocol – proper scores, skill against climatology, a hold-out intensity test, and bootstrap and multi-split robustness.

4.1 Problem set-up and notation

Let $\mathcal{I}_t \geq 0$ denote the daily standardised precipitation index on day t , and let $u > 0$ be a fixed exceedance threshold (we take $u = 1$, the value at which the index marks an extreme, and report sensitivity to u in Section 5.1). Because $\mathcal{I} = P/P_q$ is itself standardised against the training-period q th percentile (Section 3.1), u and q play distinct roles: q sets which event counts as rare in the first place (e.g. the 99th-percentile analysis of Section 3.4), while u sets how far above that level counts as extreme – e.g. $u = 1.25$ requires exceeding 125% of the q th-percentile level. A day is *extreme* when $\mathcal{I}_t > u$. Write the occurrence indicator and the excess as

$$O_t = \mathbf{1}\{\mathcal{I}_t > u\}, \quad Y_t = (\mathcal{I}_t - u) \mid \{\mathcal{I}_t > u\}, \quad (2)$$

so that $O_t \in \{0, 1\}$ is observed every day while the excess Y_t is *defined and observed only on extreme days*. This asymmetry — many occurrence observations, few intensity observations — shapes both the modelling (Section 4.2) and the separate evaluation of occurrence and intensity predictability (Section 4.4).

Each day carries a vector of covariates $\mathbf{x}_t \in \mathbb{R}^d$ that are *strictly antecedent*: functions of information available up to and including day t (autoregressive lags and rolling summaries of

the index, plus lagged ERA5 atmospheric drivers and teleconnection indices; see Section 3). Let $\mathcal{F}_t$ be the filtration generated by all information up to day t . For a forecast horizon $h \geq 1$ the target is the *predictive distribution* of $\mathcal{I}_{t+h}$ given $\mathcal{F}_t$, summarised by its conditional CDF

$$F_{t+h}(z) = \mathbb{P}(\mathcal{I}_{t+h} \leq z \mid \mathcal{F}_t), \quad z \geq 0. \quad (3)$$

Two functionals are of primary interest: the *exceedance probability* $\pi_{t+h} = \mathbb{P}(\mathcal{I}_{t+h} > u \mid \mathcal{F}_t) = 1 - F_{t+h}(u)$ and the conditional law of the excess $Y_{t+h} \mid \{\mathcal{I}_{t+h} > u\}$.

4.2 Models compared

We assess a ladder of models of increasing complexity, all sharing the same strictly antecedent covariates $\mathbf{x}_t$ and judged by the protocol of Section 4.4. The aim is not to crown one architecture but to learn *how much* of the index is predictable and *which* modelling choices matter. Models fall into two groups: *occurrence-only classifiers* (which estimate only the exceedance probability π_{t+h} , and are not scored on the full distribution) and *full-distribution models* (which output the complete CDF F_{t+h} and are additionally evaluated with the twCRPS of Section 4.4).

Baselines. *Climatology* issues the empirical distribution of the training index, the reference against which every skill score is measured; a *stationary peaks-over-threshold* (POT) model adds a Generalised Pareto tail (Davison and Smith, 1990) fitted to the training excesses but no covariate dependence.

Occurrence: logistic regression and gradient-boosted trees. For the exceedance probability $\pi_{t+h} = \mathbb{P}(\mathcal{I}_{t+h} > u \mid \mathcal{F}_t)$ we fit a regularised logistic regression on $\mathbf{x}_t$ — the natural calibrated occurrence model — and, as a strong non-linear machine-learning baseline, an isotonic-calibrated gradient-boosted-tree classifier (XGBoost) (Chen and Guestrin, 2016; Friedman, 2001), with tree scores mapped to probabilities by isotonic calibration (Zadrozny and Elkan, 2002).

Full-distribution zero-inflated regression. Rather than hurdle at the extreme threshold — which restricts intensity estimation to the few exceedances — we model the *entire* distribution of $\mathcal{I}$ using all wet days and derive the exceedance probability and tail from it. With a point mass $p_0(\mathbf{x}_t)$ at zero and a positive continuous CDF F_+ ,

$$F_{t+h}(z) = p_0 + (1 - p_0) F_+(z; \boldsymbol{\theta}(\mathbf{x}_t)) \quad (z > 0), \quad \pi_{t+h} = (1 - p_0)(1 - F_+(u)), \quad (4)$$

with $\text{logit}(p_0(\mathbf{x}_t)) = \mathbf{w}_0^\top \mathbf{x}_t$ and the distribution parameters $\boldsymbol{\theta}(\mathbf{x}_t)$, modelled through parameter-specific inverse link functions applied to linear predictors, ensuring that the required parameter constraints (e.g. positivity of scale parameters) are satisfied. The Generalised Pareto distribution is

$$H(y; \sigma, \xi) = 1 - (1 + \xi y / \sigma)_+^{-1/\xi} \quad (\xi \neq 0), \quad 1 - e^{-y/\sigma} \quad (\xi = 0), \quad y \geq 0, \quad (5)$$

with $(a)_+ = \max(a, 0)$ and ξ governing tail heaviness. For F_+ we compare three choices: (i) a *log-normal*, (ii) a *gamma*, and (iii) the *extended GPD* (eGPD) (Naveau et al., 2016) — the extreme-value-motivated choice —

$$F_+(z) = [H(z; \sigma, \xi)]^\kappa, \quad \kappa > 0, \quad (6)$$

which is GPD-tailed at high values (inheriting the Pickands–Balkema–de Haan justification of extreme-value theory; Coles, 2001) yet is a single continuous model over all positive intensities, requiring *no* threshold selection (κ shapes the lower tail).

Hurdle with a GPD tail: a Gaussian-process variant. The most flexible contender is a two-part hurdle whose exceedance intensity component is a GPD with a covariate-dependent scale parameter,

$$F_{t+h}(z) = (1 - \pi_{t+h}) + \pi_{t+h} H(z - u; \sigma_{t+h}, \xi), \quad z > u, \quad (7)$$

with $\text{logit}(\pi_{t+h}) = g_\pi(\mathbf{x}_t)$ and $\sigma_{t+h} = \log\{1 + \exp(g_\sigma(\mathbf{x}_t))\}$ driven by latent functions g_π, g_σ on which we place Gaussian-process priors — a single-layer GP with a standard automatic-relevance-determination (ARD) squared-exponential kernel (Rasmussen and Williams, 2006) acting directly on the covariates, fitted by sparse variational inference (Salimbeni and Deisenroth, 2017). We refer to it as the *GP-EVT hurdle* throughout (Table 2). A Gaussian-likelihood GP cannot represent the zero-inflation and heavy tail of $\mathcal{I}$, which motivates the EVT likelihood. Full inference details are in Appendix B.

The classical hurdle assumes occurrence and intensity are conditionally independent; physically they should be positively correlated, and coupling them could let the abundant occurrence data inform the data-starved tail. In plain terms: rather than fitting the occurrence function g_π and the intensity function g_σ independently, we let g_σ borrow statistical strength from the much better-observed g_π , in proportion to an estimated correlation between the two. We realise this by giving (g_π, g_σ) a multi-output GP prior with an intrinsic coregionalisation (Bonilla et al., 2008; Álvarez et al., 2012), $\text{Cov}(g_a(\mathbf{x}), g_b(\mathbf{x}')) = B_{ab} k_\theta(\mathbf{x}, \mathbf{x}')$, whose off-diagonal yields an interpretable borrowing-strength correlation $\rho = B_{\pi\sigma} / \sqrt{B_{\pi\pi} B_{\sigma\sigma}}$ ($\rho = 0$ recovers the independent hurdle). Because the excess is observed only on extreme days ρ is weakly identified; as we report in Supplementary Section S4, on this data the coupling turns out to be empirically inert — neither helping nor hurting the forecasts — so we retain it only as a harmless nested generalisation of the independent hurdle. This construction is related to correlated two-part models in biostatistics and econometrics (Olsen and Schafer, 2001; Heckman, 1979).

Implementation. Logistic regression uses L_2 regularisation ($C = 1$); the L_1 feature screen uses $C = 0.3$; XGBoost uses 300 trees, depth 4, learning rate 0.05. These are standard off-the-shelf settings, fixed *a priori* and not tuned by cross-validation; a reviewer could ask whether lightly tuning C would change the ordering, but $C = 1$ is itself a weak, unremarkable amount of regularisation and the occurrence models are already essentially tied at the top, so we consider this unlikely to matter. The distributional regressions place linear predictors on the zero-inflation probability and the positive-part parameters and are fitted by maximum likelihood. The GP-EVT hurdle uses a sparse variational GP ($M = 64$ inducing points) with the coregionalised prior above and a standard ARD kernel; a sensitivity sweep over inducing points ($M \in \{32, 64, 128\}$) confirms the choice is not load-bearing (Section 5.1). Gaussian-process results are averaged over three random seeds; conformal intervals target 90% coverage.

4.3 Tail-conformal calibration

Any of the models above yields a model-based predictive tail, but under misspecification its quantiles need not be calibrated. As one calibration option we assess, we wrap a model in a conformal layer (Vovk et al., 2005; Romano et al., 2019) that supplies distribution-free coverage, with two adaptations for our setting. *Tail targeting*: at the very high confidence levels relevant to extremes the required nonconformity quantile can exceed what a finite calibration set resolves (the empirical quantile saturates); when this happens we extrapolate the upper tail with a GPD fitted to the top 20% of calibration scores (a data-adaptive, not fixed-probability, exceedance threshold), following the extreme-conformal construction of Pasche et al. (2025). *Temporal adaptation*: because daily extremes are not exchangeable, we use adaptive conformal inference (Gibbs and Candès, 2021), updating the level online. Here α_t is the working miscoverage level used to build the interval C_t issued on day t : it starts at the nominal level α and is then adjusted

up or down after each test day depending on whether the previous interval covered the outcome, so that realised coverage tracks the nominal target even as the process drifts,

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t), \quad \text{err}_t = \mathbf{1}\{\mathcal{I}_t \notin C_t\}, \quad (8)$$

with C_t the issued interval and step size $\gamma = 0.02$, over an *expanding* calibration pool: starting from the held-out calibration set, each test-day nonconformity score is appended to the pool after it is scored, so the reference distribution for C_t grows through the test period rather than sliding over a fixed-width window. Under (8) with bounded scores the long-run coverage converges to $1 - \alpha$ irrespective of the data-generating process (Gibbs and Candès, 2021). We stress that this guarantee concerns *coverage*, not predictive quality: as Section 5.2 shows, a conformally calibrated model can still be worse than climatology on proper scores, so coverage alone is an insufficient indicator and we never report it as evidence of a good forecast.

4.4 Evaluation protocol

Because the predictability of extremes is intrinsically limited, evaluation must be both *proper* and *robust*: it must reward calibrated, sharp forecasts (maximising sharpness subject to calibration; Gneiting et al., 2007) and must resist the artefacts — single-run noise, quantile selection — that easily manufacture apparent skill in a low-signal regime. We therefore adopt the following protocol. Its individual ingredients are standard; our contribution is their disciplined integration into a single artefact-resistant assessment for a low-signal, rare-event regime — in particular, pairing calibration-aware proper scores with skill against climatology and multi-split robustness, and demonstrating that distribution-free conformal coverage, though guaranteed, is not on its own evidence of forecast quality.

Occurrence: Brier decomposition and skill. We assess the exceedance probability with the Brier score, whose reliability/resolution/uncertainty decomposition (Murphy, 1973),

$$\text{BS} = \underbrace{\frac{1}{n} \sum_k n_k (\bar{p}_k - \bar{o}_k)^2}_{\text{reliability}} - \underbrace{\frac{1}{n} \sum_k n_k (\bar{o}_k - \bar{o})^2}_{\text{resolution}} + \underbrace{\bar{o}(1 - \bar{o})}_{\text{uncertainty}}, \quad (9)$$

groups forecasts into bins k (with n_k cases, mean forecast $\bar{p}_k$, and observed frequency $\bar{o}_k$, and overall base rate $\bar{o}$): the first term is reliability (calibration), the second resolution (informativeness), and the third the irreducible uncertainty common to all forecasts. We report the Brier skill score against climatology together with the resolution term, since a calibrated-but-uninformative forecast (e.g. climatology) has zero resolution: a useful model must therefore clear the bar of positive resolution while retaining good reliability.

Whole distribution and tail: (tw)CRPS. We score the predictive distribution with the CRPS (Gneiting and Raftery, 2007) and its threshold-weighted form (twCRPS) (Gneiting and Ranjan, 2011),

$$\text{twCRPS}(F_{t+h}, \mathcal{I}_{t+h}) = \int_{\mathbb{R}} (F_{t+h}(z) - \mathbf{1}\{\mathcal{I}_{t+h} \leq z\})^2 w(z) dz, \quad w(z) = \mathbf{1}\{z > \tau\}, \quad (10)$$

which remains proper while emphasising the tail ($\tau = u$). With $\tau = u$ the sub-threshold distribution is irrelevant, so the tail score depends only on the exceedance probability and the modelled excess.

Because the twCRPS over all days is dominated by the many non-extreme days, we additionally isolate intensity by scoring the *excess* on observed extreme days: a CRPS of the excess (equivalently, the integral of the pinball loss over quantile levels) relative to the climatological excess distribution. The excess model is a quantile regression (Koenker and Bassett, 1978) whose regularisation is chosen by cross-validation on the training set only, so the test set never informs model selection.

Robustness. Skill scores are computed against a climatological reference (the empirical training distribution), and every estimate carries a moving-block bootstrap (Künsch, 1989) confidence interval (1,000–2,000 resamples, blocks of ten extreme days to respect short-range temporal dependence). The intensity result is further replicated across the four forward-only, expanding-window splits of Section 3.4 *and across the two cities*.

The block length deserves its own check. It is not tightly identified by the short-range autocorrelation of Figure 1d alone, and the smallest split has only 96 test extremes in London (roughly ten blocks of ten). We therefore recomputed the interval at block lengths of 5 and 20 at that split: the 90% CI moves by at most ± 0.003 around the reported $[+0.006, +0.059]$ in London and $[+0.019, +0.073]$ in Paris (Table 3), so the block-10 choice is not load-bearing in the smallest-sample case.

Finally, per-feature significance is controlled for multiplicity by the Benjamini–Hochberg false-discovery rate; feature importance is quantified by stability selection (selection frequency under subsampling); and the GP–EVT hurdle is probed by a sensitivity sweep over inducing points. Discrimination metrics (precision–recall AUC, PR-AUC) are reported only as secondary context, and conformal coverage is reported but, per Section 4.3, never as evidence of predictive quality.

5 Results

We evaluate every model on the 95th-percentile index of *both cities* by the proper scores of Section 4.4 against a climatological reference, with stochastic results seed-averaged and intensity skills carrying moving-block bootstrap confidence intervals. Beyond the main bake-off (Section 5.1) we report a feature screen (with false-discovery control and stability selection) identifying which covariates carry occurrence and intensity signal; the hold-out intensity test across four expanding-window splits; a GP inducing-point sensitivity sweep; a seed-averaged horizon sweep $h \in \{1, 3, 7, 14, 28\}$; and a synthetic study probing the dependent-hurdle coupling (Supplementary). Throughout, we ask whether conclusions hold across the two contrasting cities.

5.1 A GP–EVT hurdle wins the tail; distribution choice matters

Table 2 reports the bake-off, and the same ordering holds in *both* cities. First, among the full-distribution models the Gaussian-process–EVT hurdle is the most skilful on the tail: it is the only such model with positive threshold-weighted CRPS skill in data-scarce London (+0.032) and clearly the best in Paris (+0.094), and in both cities a 90% moving-block bootstrap interval over the test period excludes zero ($[+0.010, +0.057]$ and $[+0.049, +0.134]$ respectively). On occurrence its Brier skill score (+0.037 in London, +0.123 in Paris; bootstrap intervals $[+0.020, +0.058]$ and $[+0.073, +0.170]$) matches the best inexpensive occurrence models — a calibrated logistic regression and gradient-boosted trees (XGBoost). No model is far ahead on occurrence, so the GP hurdle earns its place through the tail rather than through discrimination. Second, among the simpler distributional regressions the choice of response distribution matters more than the choice of model: a zero-inflated log-normal is substantially miscalibrated (twCRPS skill -2.3 in London), whereas the light-tailed gamma and the threshold-free extended GPD are well behaved; the index tail is lighter than a log-normal or a low-threshold Generalised Pareto fit assumes. Third, modelling the *full* zero-inflated distribution and deriving the exceedance probability from it beats hurdling at the extreme threshold, which starves the tail of data. Quantitatively everything is *stronger in Paris* (e.g. logistic Brier skill +0.117 versus London’s +0.038), reflecting its larger, more convective extreme sample; indeed in Paris the extended GPD too beats climatology on the full-predictive tail score (twCRPS +0.052), which it could not in data-scarcer London.

Is the GP result robust to its settings? Yes. A sensitivity sweep over inducing points ($M \in \{32, 64, 128\}$) leaves the GP-EVT hurdle’s skill essentially unchanged in both cities — Brier skill +0.03 to +0.04 in London and +0.12 in Paris, with threshold-weighted CRPS skill positive throughout — so the $M = 64$ choice is not load-bearing. The hurdle uses a standard ARD kernel with no cross-validated tuning, so its tail advantage comes from a well-matched EVT likelihood and a modest, standard Gaussian process rather than from any delicate configuration; it is about as flexible as this low-signal problem can bear.

Sensitivity to the exceedance threshold u . Re-fitting the winning occurrence model (logistic +ERA5) at $u \in \{0.75, 1.00, 1.25\}$ — i.e. marking a day extreme at 75%, 100%, and 125% of the q th-percentile level — leaves the qualitative finding unchanged in both cities: the Brier skill score stays clearly positive and of comparable magnitude at every u (London +0.055, +0.038, +0.028; Paris +0.103, +0.117, +0.101). In London skill rises modestly as u falls, as expected — a lower threshold admits more extreme days and so more data to fit — while Paris peaks at $u = 1$; neither city shows the skill collapsing or changing sign near the working threshold. This is reassuring on two counts: predictability is not an artefact of the particular threshold $u = 1$, and it is not a knife-edge effect that appears at one value and vanishes at neighbouring ones.

5.2 Conformal coverage does not imply forecast quality

The tail-conformal layer attains its nominal 0.90 coverage on the real series across every predictor set and horizon (Figure 4b). But this coverage is *guaranteed by construction* for *any* base model, so it is uninformative about predictive quality: a zero-inflated log-normal, for instance, is worse than climatology on the proper scores (Table 2, twCRPS skill -2.3 in London) yet would attain the same nominal conformal coverage. The reliability diagram (Figure 2) makes the point directly: the zero-inflated log-normal’s raw exceedance probabilities are badly miscalibrated (systematically overconfident), whereas the logistic regression and the GP-EVT hurdle both track the diagonal and spread their predictions (calibrated and informative). Conformal coverage should therefore not be reported as evidence of a good probabilistic forecast — a cautionary point in its own right.

5.3 Predictability of occurrence

Occurrence is weakly but genuinely predictable in both cities: the calibrated logistic regression improves on climatology with a Brier skill of +0.038 in London and +0.117 in the more convective Paris (Table 2), with positive resolution. An L_1 -regularised logistic identifies physically sensible antecedent predictors — led by the K-index of convective instability, the 500 hPa geopotential, integrated vapour transport, low-level humidity, and season — and stability selection confirms their robustness: the K-index and the most recent index lag are selected in essentially every training subsample (full ranking, both cities: Supplementary Table S2). Skill nonetheless decays toward the prevalence floor as the forecast horizon grows (Figure 4a; numerical BSS by horizon, both cities: Supplementary Table S3), consistent with the limited day-scale persistence of these extremes. At the rarer 99th-percentile threshold (London), where the scarce data preclude the held-out intensity test, the qualitative picture is unchanged (Supplementary Table S1).

5.4 Predictability of intensity

Whether the *magnitude* of an extreme is predictable is the more delicate question, and the answer is a qualified yes — in both cities. On the extreme days, many antecedent features correlate with the excess far beyond chance, and the association survives multiplicity control: 21 of 49 features in London and 27 of 49 in Paris pass a Benjamini–Hochberg false-discovery threshold (versus ~ 2 expected), led by convective inhibition and CAPE in both. To test whether this

signal generalises out of sample, we fit quantile regressions of the excess on held-out extreme days (regularisation chosen by cross-validation on the training set only) and score them by the CRPS of the excess against the climatological excess distribution. Computed over a fine quantile grid on the primary split, the CRPS skill is small but its moving-block bootstrap interval *excludes zero* in each city (London +0.014, [+0.009, +0.019]; Paris +0.036, [+0.005, +0.074]). As a robustness check we recompute it over a coarser quantile grid across four successive expanding-window splits per city: the skill stays positive throughout — every interval excluding zero (though in the two smallest windows the interval only marginally clears zero, $P(> 0) = 0.97\text{--}0.99$), with primary-split (2009–2018) values of +0.035 in each city (Table 3, Figure 3). The signal is concentrated in the upper tail (the largest events) and is carried by antecedent convective instability (CAPE/CIN); median magnitude is indistinguishable from climatology. The Paris point estimates run somewhat higher than London’s, but their confidence intervals overlap, so we read the intensity signal as robustly *replicated* across the two cities rather than decisively stronger in either.

5.5 Coupling the hurdle: a null result

We also tested whether *coupling* the occurrence and intensity halves of the hurdle through coregionalisation (Section 4.2) improves forecasts, and found it weakly identified and empirically inert at this data volume — switching the coupling off leaves the bake-off skill unchanged — so we retain it only as a harmless nested generalisation of the independent hurdle; the full synthetic-recovery, misspecification, and real-data $\hat{\rho}$ analyses are in Supplementary Section S4.

5.6 Skill decays with horizon

Both occurrence skill and discrimination decay toward the climatological floor as the forecast horizon grows (Figure 4a), consistent with the limited day-scale persistence of London extremes; conformal coverage, being guaranteed by construction, is flat across horizons (Figure 4b). The +ERA5 drivers lift discrimination clearly at $h = 1$ (PR-AUC 0.118 versus 0.076 for index history alone) but the edge fades into seed noise by $h = 3$ — a reminder that the predictability documented above is a short-lead effect, weak and easily masked unless assessed with seed-averaging and proper scores rather than single runs. The same decay and flat coverage hold in Paris (Figure 4c,d).

Synthesis. Across the bake-off, a GP–EVT hurdle is the strongest full-distribution forecaster while simple calibrated models match it on occurrence, and the response distribution matters as much as the model class (Section 5.1); conformal coverage holds but is, on its own, uninformative about forecast quality (Section 5.2). The substantive finding is a small but *rigorously established* predictability: occurrence is weakly predictable from antecedent convective-instability and circulation indices (Section 5.3), and the magnitude of the largest events is weakly predictable from antecedent convective instability, with a CRPS skill over climatology that excludes zero across four successive splits (Section 5.4). The optional dependent-hurdle coupling, tested separately, proves empirically inert at this data volume (Section 5.5). Every one of these conclusions holds in *both* London and Paris, and is generally more pronounced in the more convective Paris — evidence that they reflect the meteorology of precipitation extremes rather than the idiosyncrasies of one city.

6 Discussion

Predictability is real but small, and physically interpretable. The central result is not strong forecast skill but a careful measurement of how much skill is genuinely present: in both cities, the occurrence of a daily precipitation extreme and the magnitude of the largest events carry

weak antecedent predictability, on the order of a few percent of CRPS skill over climatology. Such effect sizes are modest but not unusual for precipitation at these lead times — sub-seasonal precipitation forecast skill is itself typically small — and their practical value lies less in a sharp point forecast than in *trustworthy, calibrated* probabilities, plus a slim but real edge for anticipating the most intense events. That the signal is led by convective-instability indices (the K-index for occurrence; CAPE and convective inhibition for intensity) is reassuring: it is the thermodynamics one would expect to govern intense convective rainfall, it argues against the skill being an artefact, and it is precisely why the effects are *larger in the more convective Paris*. The median magnitude of an extreme, by contrast, is indistinguishable from climatology: once an exceedance occurs, its size is largely unpredictable except in the upper tail. One nuance reinforces the data-dependence of these statements: in data-scarcer London no model beats the empirical climatological tail on the full-predictive score, but in Paris — with half again as many extremes — the extended GPD (Davison and Smith, 1990; Coles, 2001) does, so whether the tail is climatology-bound is a sample-size effect rather than a universal law.

Three methodological lessons are worth emphasising. First, *a compact, well-matched model was enough*: in both cities a single-layer Gaussian-process hurdle with a standard kernel and an EVT tail was the strongest full-distribution forecaster on the tail, yet on occurrence it was matched by an inexpensive logistic regression and gradient-boosted trees — so added model flexibility yielded little benefit, and a modest Gaussian process is about as flexible as this low-signal problem can bear. Gaussian processes would offer greater benefit through structure that the single-site temporal problem does not provide (notably spatial structure across the grid of cells). Second, *the response distribution dominates*: a log-normal tail was substantially miscalibrated, and hurdling at the extreme threshold starved the tail of data; modelling the full zero-inflated distribution with a threshold-free extended GPD was far better. Third, and most relevant to the wider literature, *conformal coverage is insufficient as a stand-alone indicator of forecast quality* (Gneiting and Raftery, 2007; Lerch et al., 2017): it is attained by construction even by a model worse than climatology on proper scores (the zero-inflated log-normal here). This is a known property of coverage, but our experiments make it concrete — reporting coverage without a proper-score comparison can create a false impression of a good probabilistic forecast.

We report occurrence and intensity separately for interpretability, but they are not disconnected: they are the point-mass and tail components of a single hurdle predictive distribution, $F(z) = (1 - \pi) + \pi H(z - u)$, with occurrence probability $\pi(\mathbf{x})$ and a covariate-dependent excess law H . Because the (unstructured) hurdle likelihood factorises in these parameters, fitting the two components jointly is *equivalent* to fitting them separately — they are orthogonal pieces of one distribution rather than two ad hoc models — and a single threshold-weighted CRPS over F decomposes additively into an occurrence and an intensity contribution. Fitting this unified hurdle directly and scoring it end-to-end (Appendix C) recovers exactly the two-part story: in both cities the overall skill splits into a modest occurrence term and a small but unusually stable intensity term (its bootstrap interval excludes zero), with negligible interaction. The only way to make the joint fit differ from the separate one is to *couple* the components through shared latent structure — the dependent GP hurdle. We tested this coupling and found it empirically inert at this data volume (Supplementary Section S4): the coupled and independent hurdles score identically, and ρ is too weakly identified on real data to estimate reliably. The separable (independent) treatment is therefore the well-identified reference, and the coupling is retained only as a harmless nested generalisation that reduces to it as $\rho \rightarrow 0$; whether a richer dataset could make borrowing strength pay is left open.

Several apparent results did not survive scrutiny: a single-seed driver benefit and an apparent long-horizon gain both evaporated under seed-averaging, and an across-the-board intensity skill shrank to an upper-tail effect once regularisation was chosen without peeking at the test set. The predictability we report is what remained after seed-averaging, proper scoring, cross-validated model selection, bootstrap intervals (Künsch, 1989), and multi-split robustness. In a low-signal

regime these safeguards are not optional – they are the difference between a finding and an artefact.

6.1 Limitations

Four limitations qualify these conclusions. First, the assessment covers only two cities. Although this is a step beyond a single site, London and Paris share a climate zone, a downscaled reanalysis product (ERA5-2km), and a common data provider, so they are not fully independent replicates and shared reanalysis biases could inflate their apparent skill together; replication in a climatically distinct setting is the most direct next step. Second, the index is itself an ERA5-derived precipitation product, which raises a circularity concern; we mitigate it by excluding all precipitation fields from the predictors and lagging every driver, so that only antecedent dynamical and thermodynamic state informs the forecast, though a residual dependence through the reanalysis cannot be excluded entirely. Third, the study uses a perfect-prognosis design based on antecedent reanalysis fields rather than operational forecasts, and the intensity analysis conditions on observed exceedances; it therefore isolates predictability rather than delivering an end-to-end warning, and moving to numerical-weather-prediction drivers is a natural extension. Fourth, we treat the 1989–2018 climatological reference as stationary; given that projections indicate a rising trend in extreme rainfall over this period (IPCC, 2021), a slowly time-varying baseline could sharpen the reported skill scores, whose likely direction is a modest downward bias in later test years.

7 Conclusion

We set out to measure, rather than merely to model, the predictability of rare precipitation extremes using an evaluation protocol designed for low-signal settings. The answer to how such predictability should be assessed is methodological: calibration-aware comparisons of proper scores against climatological benchmarks reveal differences among models, and between occurrence and full-distribution skill, that conformal coverage alone cannot. Although all models achieved nominal coverage, only proper-score comparisons identified which forecasts genuinely contained predictive information.

The answer to how predictable the extremes themselves are is more qualified: there is modest but measurable predictability. The occurrence and magnitude of the largest events are weakly but consistently informed by antecedent convective instability, with a compact Gaussian-process extreme-value hurdle model providing the strongest full-distribution forecasts in the tail, while simpler calibrated models perform similarly for occurrence alone. The consistency of these findings across the more convective Paris and the maritime London suggests that the identified signal is not specific to a single location, while its stronger expression in Paris highlights the influence of convective regimes. For the flood managers and insurers who motivated this study, antecedent conditions therefore provide a small but genuine and honestly calibrated source of predictive information.

Natural extensions are those in which additional signal, and a stronger role for structured Gaussian-process models, are most plausible: expanding to more cities to characterise predictability across climates, introducing spatial dependence across grids of cells, incorporating operational drivers from numerical weather prediction rather than reanalysis, and considering sub-seasonal aggregations where slowly varying predictors may contribute. Whether predictability increases materially in these settings remains, in the spirit of this paper, a question to be answered by proper scores and uncertainty quantification rather than asserted.

Supporting Information

Additional information and supporting material for this article is available online at the journal’s website.

Data and code availability

The underlying data are publicly available: the London and Paris precipitation-extreme indices derive from the Copernicus Climate Data Store product *Extreme precipitation risk indicators for Europe and European cities (1950–2019)*, and the atmospheric predictors from ERA5 (Copernicus CDS, `reanalysis-era5-single-levels`; and the WeatherBench2 regrid-ded ERA5 archive) and from the NOAA teleconnection indices. The code accompanying this paper — implementations of all models (logistic, gradient-boosted trees, the zero-inflated log-normal/gamma/eGPD distributional regressions, and the GP–EVT hurdle), the full evaluation protocol (proper scores, the hold-out intensity test, moving-block bootstrap and multi-split robustness), the data-retrieval and feature-processing scripts, and the scripts that produce every figure and table — together with the processed daily feature tables for both cities, are publicly available at <https://github.com/olatunjijohnson/precip-extremes-predictability>. Following journal policy, the processed data files will additionally be deposited on figshare with a citable DOI upon acceptance.

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Conflict of interest

The authors declare no conflict of interest.

Appendix

A Data and predictors

Response. The standardised daily precipitation-extreme index for London and Paris (Section 3), derived from the ERA5-2km downscaled reanalysis; we use the daily maximum over each city’s grid cells (53 for London, 157 for Paris), with an extreme defined by $\mathcal{I} > 1$. We analyse the 95th-percentile standardisation for both cities and, for London only, the rarer 99th percentile (Supplementary Material).

Antecedent predictors. All predictors are functions of information up to the forecast origin t (a horizon- h forecast of day $t + h$ uses day- t information only); no precipitation field is used, since the response is itself an ERA5-derived precipitation product.

- **Index history:** lags $\mathcal{I}_{t-1}, \dots, \mathcal{I}_{t-14}$; rolling mean over the previous 7 and 30 days; rolling maximum over 7 days; count of exceedances in the previous 30 days; and sin / cos harmonics of day-of-year.
- **ERA5 circulation and moisture** (WeatherBench2, 1.5°, daily means over a city-centred synoptic box, $\sim 15^\circ \times 30^\circ$ — e.g. 45–60°N, 20°W–10°E for London, shifted south-east for Paris): mean sea-level pressure and its meridional/zonal gradients, total-column water vapour,

2 m temperature, 10 m wind components and speed, geopotential at 500 and 850 hPa, and specific humidity at 850 hPa.

- **ERA5 convective instability** (Copernicus CDS, `reanalysis-era5-single-levels`, 0.25°, daily mean and maximum over the box): convective available potential energy (CAPE), convective inhibition (CIN), the K-index, the Total-Totals index, total-column water vapour, the vertical-integral eastward/northward water-vapour-flux components and their magnitude (IVT).
- **Teleconnection indices** (monthly, broadcast to daily): North Atlantic Oscillation and Arctic Oscillation (NOAA CPC), and Niño 3.4 (NOAA PSL).

Retrieval. WeatherBench2 fields were read directly from the public Google Cloud zarr store with `xarray`; the convective-instability fields were retrieved from the Copernicus CDS via `cdsapi` with server-side spatial subsetting to the box, 6-hourly sampling aggregated to daily statistics; teleconnection indices were downloaded from NOAA. Every driver is forward-filled to daily resolution where needed and lagged by the forecast horizon. Processing code and the cached daily feature table accompany the paper.

B The Gaussian-process extreme-value hurdle: inference

We give the inference for the Gaussian-process-EVT hurdle (Section 4.2). The parameters depend on covariates through latent functions, $\pi = \text{sigmoid}(g_\pi(\mathbf{x}_t))$, $\sigma = \text{softplus}(g_\sigma(\mathbf{x}_t))$, and a constant shape ξ_0 (covariate-dependent ξ is an optional extension). For horizon h the per-day likelihood factorises into occurrence and (on extreme days) intensity terms,

$$p(O_t, \mathcal{I}_t \mid \mathbf{g}(\mathbf{x}_{t-h})) = \pi^{O_t} (1 - \pi)^{1-O_t} [h_{\text{GPD}}(\mathcal{I}_t - u; \sigma, \xi_0)]^{O_t} [f^{\text{blw}}(\mathcal{I}_t)]^{1-O_t}, \quad (11)$$

with h_{GPD} the GPD density and f^{blw} a zero-inflated positive density for the sub-threshold regime.

Sparse variational inference. We place GP priors on $\mathbf{g} = (g_\pi, g_\sigma)$ and introduce $M \ll N$ inducing variables $\mathbf{u} = \mathbf{g}(\mathbf{Z})$ with whitened variational posterior $q(\mathbf{u}) = \mathcal{N}(\mathbf{m}, \mathbf{S})$. The evidence lower bound is

$$\mathcal{L} = \sum_t \mathbb{E}_{q(\mathbf{g}(\mathbf{x}_{t-h}))} [\log p(O_t, \mathcal{I}_t \mid \mathbf{g}(\mathbf{x}_{t-h}))] - \text{KL}(q(\mathbf{u}) \parallel p(\mathbf{u})), \quad (12)$$

where KL is the Kullback–Leibler divergence and the expectation is computed by Monte Carlo over the marginal $q(\mathbf{g}(\mathbf{x}_{t-h}))$. Inducing-point inference reduces cost to $\mathcal{O}(NM^2)$; the kernel is a standard automatic-relevance-determination squared-exponential kernel acting directly on the covariates, fitted by (doubly-stochastic) variational inference (Salimbeni and Deisenroth, 2017). Hyperparameters, inducing inputs/distribution, and ξ_0 are learned jointly by maximising (12).

Coregionalised (dependent) hurdle. We couple the two latents through a linear model of coregionalisation: with Q shared base GPs $f_q \sim \mathcal{GP}(0, k_\theta)$, $g_a(\mathbf{x}) = \sum_q W_{aq} f_q(\mathbf{x})$, so $\text{Cov}(g_a, g_b) = B_{ab} k_\theta$ with $\mathbf{B} = \mathbf{W}\mathbf{W}^\top$. The correlation $\rho = B_{\pi\sigma} / \sqrt{B_{\pi\pi} B_{\sigma\sigma}}$ is the borrowing-strength parameter; a diagonal $\mathbf{W}$ recovers the independent hurdle. Because the excess is observed only on extreme days, ρ is weakly identified, so we use a weakly-informative prior shrinking $\rho \rightarrow 0$; we examine its recovery on synthetic data and its (null) effect on forecasts in Supplementary Section S4.

Tail-conformal layer. For an interval at level $1 - \alpha$ we use a conformalised-quantile-regression nonconformity score on the model’s predictive quantiles, $E_t = \max\{\hat{z}_{\alpha/2} - \mathcal{I}_t, \mathcal{I}_t - \hat{z}_{1-\alpha/2}\}$; when the required score quantile exceeds what the calibration set resolves we extrapolate with a GPD fitted to the top 20% of the upper calibration scores (Pasche et al., 2025), and we adapt the level online by (8) (step size $\gamma = 0.02$) over an expanding calibration pool that accumulates test-day scores as the walk-forward evaluation proceeds.

C Unified hurdle: one predictive distribution, decomposed

To verify that the separately reported occurrence and intensity analyses are the two components of one inferential object (Section 6), we fit a single hurdle predictive distribution directly,

$$F(z \mid \mathbf{x}) = \begin{cases} 1 - \pi(\mathbf{x}) & z \leq u, \\ (1 - \pi(\mathbf{x})) + \pi(\mathbf{x}) H(z - u; \sigma(\mathbf{x}), \xi) & z > u, \end{cases}$$

with $\pi(\mathbf{x}) = \text{sigmoid}(\mathbf{a}^\top \mathbf{x})$, $\sigma(\mathbf{x}) = \text{softplus}(\mathbf{b}^\top \mathbf{x})$, H the GPD excess law and ξ a covariate-free tail index, fitting all parameters by a *single* joint hurdle log-likelihood. Because that likelihood factorises, this is numerically equivalent to the separate fits; the point is the *evaluation*. We score the whole F with one threshold-weighted CRPS at $\tau = u$ and attribute its skill over a parametric climatology (constant base rate $\bar{\pi}$ plus a stationary GPD tail) by swapping one component at a time: S_{occ} uses $\pi(\mathbf{x})$ with the climatological tail, S_{int} uses $\bar{\pi}$ with $\sigma(\mathbf{x})$, and S_{full} uses both; the decomposition is additive up to a small interaction. Intervals are moving-block bootstrap (90%, block = 10 days, seed-averaged over three fits); the resulting decomposition is reported in Table 4.

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Tables and Figures

Table 1: Summary characteristics of the 95th-percentile precipitation-extreme index (1989–2018; $N = 10,957$ days) for the two cities. Extreme days are those with $\mathcal{I} > 1$. Paris is more convective: more cells, more extremes, a heavier tail.

	London (maritime)	Paris (continental)
Grid cells	53	157
Extreme days (total)	417	607
Extreme rate (%)	3.81	5.54
train 1989–2008 (7,305 d)	287 (3.93%)	396 (5.42%)
test 2009–2018 (3,652 d)	130 (3.56%)	211 (5.78%)
Days equal to zero (%)	25.5	28.7
Median index	0.015	0.012
Maximum index	6.46	8.91
Peak season	July–August	

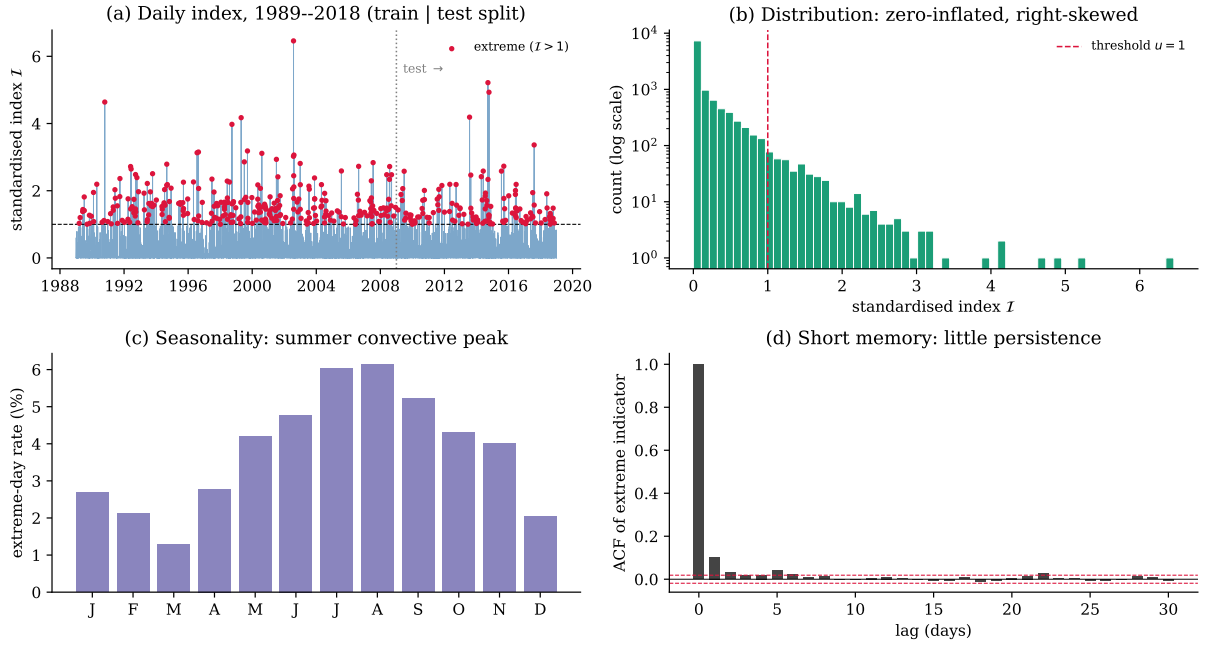


Figure 1: The London 95th-percentile precipitation-extreme index, 1989–2018 (Paris is qualitatively similar; see Supplementary Figure S1). **(a)** Daily series with extreme days ($\mathcal{I} > 1$) marked and the train/test split; **(b)** the distribution is zero-inflated ($\sim$ a quarter of days are exactly zero) and strongly right-skewed (log count); **(c)** extremes peak in summer (convective rainfall); **(d)** the extreme indicator has very short memory — the autocorrelation is negligible beyond a day or two (dashed: 95% band) — which caps how much an autoregressive forecaster can achieve.

Table 2: Model bake-off at the 95th percentile, $h = 1$ (seed-averaged), for both cities. Brier skill score (BSS) and twCRPS skill are relative to climatology, so a score of zero means climatology-level skill, positive is better, and negative is worse than climatology. Logistic and XGBoost are occurrence-only (no twCRPS). ZI denotes the full-distribution zero-inflated regressions of Section 4.2. The GP-EVT hurdle is the strongest full-distribution model on the tail in both cities. A dash (—) marks a model with no full predictive distribution (occurrence-only). The zero-inflated log-normal’s twCRPS (≈ -2.3 in London) reflects a calibration failure and is off the scale of the other entries.

Model	London		Paris	
	BSS	twCRPS	BSS	twCRPS
Climatology / stationary POT	0.000	0.000	0.000	0.000
Logistic (+ERA5)	+0.038	—	+0.117	—
XGBoost (+ERA5)	+0.034	—	+0.101	—
ZI gamma (+ERA5)	+0.008	−0.06	+0.090	+0.046
ZI extended-GPD (+ERA5)	+0.012	−0.045	+0.093	+0.052
ZI log-normal (+ERA5)	−0.29	−2.27	−0.04	−0.76
GP-EVT hurdle (+ERA5)	+0.037	+0.032	+0.123	+0.094

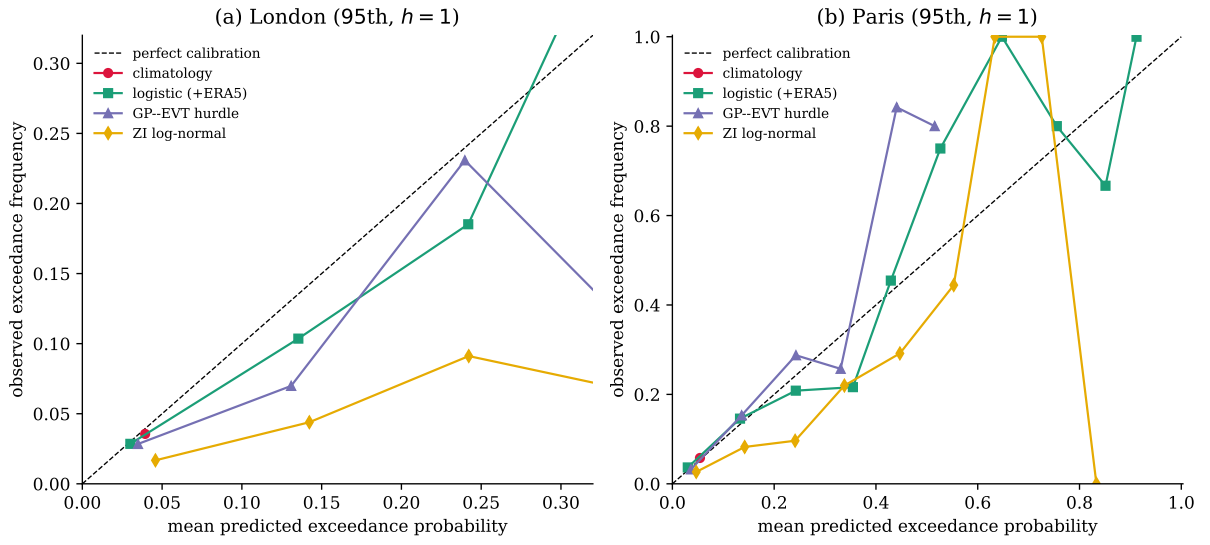


Figure 2: Reliability diagrams for the exceedance probability at the 95th percentile ($h=1$) for **(a)** London and **(b)** Paris. In both cities the logistic regression and the GP-EVT hurdle track the diagonal (calibrated) and spread their predictions (informative); the zero-inflated log-normal is systematically overconfident (its predicted exceedance probabilities exceed the observed frequencies), despite attaining nominal conformal coverage by construction. Climatology is a single point at the base rate.

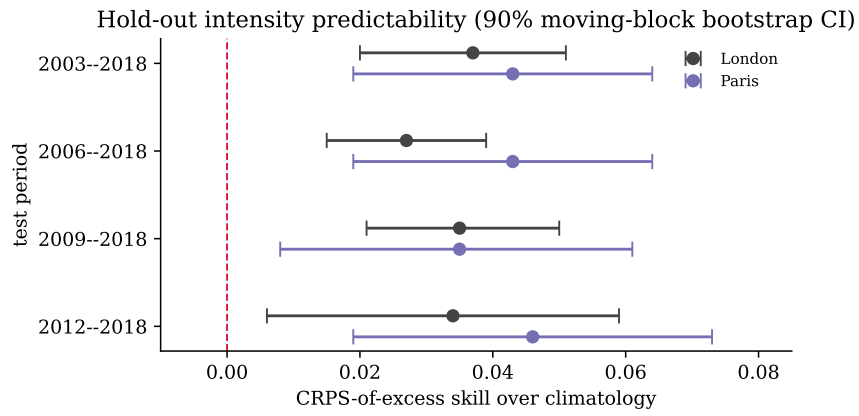


Figure 3: Hold-out intensity predictability for both cities: CRPS-of-excess skill over climatology, with 90% moving-block bootstrap intervals, across four forward-only expanding-window train/test splits. The skill is small but its interval excludes zero in every split and city; Paris point estimates run somewhat higher than London's, though the intervals overlap.

Table 3: Hold-out intensity predictability in both cities: CRPS-of-excess skill over climatology across forward-only expanding-window splits, with 90% moving-block bootstrap intervals. Every interval excludes zero; Paris point estimates run somewhat higher than London’s, though the intervals overlap.

Test period	# test ext.	CRPS skill	90% CI	P(> 0)
<i>London</i>				
2003–2018	213	+0.037	[+0.020, +0.051]	1.00
2006–2018	175	+0.027	[+0.015, +0.039]	1.00
2009–2018	130	+0.035	[+0.021, +0.050]	1.00
2012–2018	96	+0.034	[+0.006, +0.059]	0.97
<i>Paris</i>				
2003–2018	329	+0.043	[+0.019, +0.064]	1.00
2006–2018	275	+0.043	[+0.019, +0.064]	1.00
2009–2018	211	+0.035	[+0.008, +0.061]	0.99
2012–2018	164	+0.046	[+0.019, +0.073]	0.99

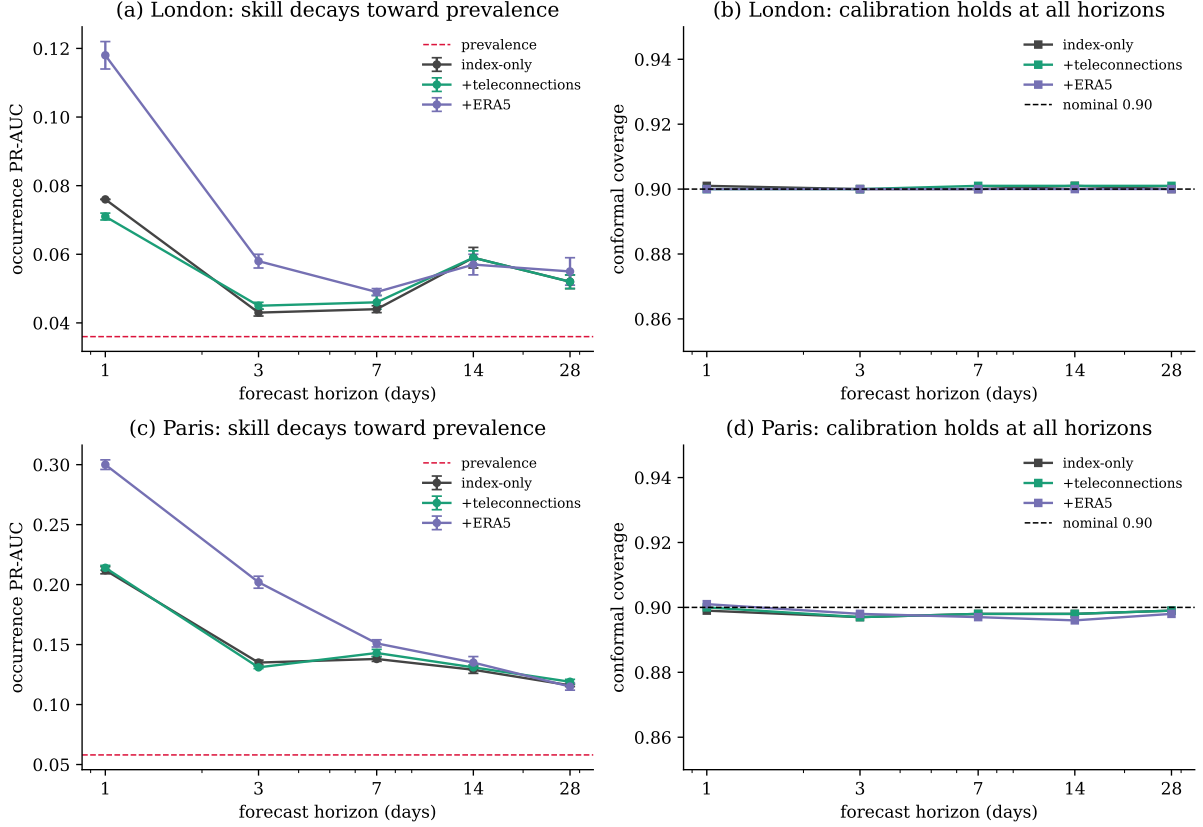


Figure 4: Seed-averaged horizon sweep on the 95th-percentile index (GP-EVT hurdle) for London (a,b) and Paris (c,d). In panels (a,c) the occurrence PR-AUC decays toward the prevalence floor (dashed) as the forecast horizon grows; the +ERA5 driver gain is clearest at $h = 1$ and fades into seed noise by $h = 3$. In panels (b,d) the conformal coverage stays at the nominal 0.90 across all horizons and predictor sets. Error bars / markers are means over three seeds. The same decay and flat coverage hold in both cities.

Table 4: The single hurdle distribution, scored end-to-end, decomposes into the two findings reported in the main text: a modest occurrence term and a small but stable intensity term (its bootstrap interval excludes zero in both cities), with negligible interaction. The unified occurrence probability reproduces the standalone logistic skill, confirming the components are orthogonal pieces of one distribution. The intensity contribution is small because it is averaged over all test days; restricted to the $\sim 3\text{--}6\%$ of days that are extreme it matches the isolated CRPS-of-excess skill of Section 5.4.

twCRPS skill vs climatology	London	Paris
S_{full} (occurrence + intensity)	+0.030 [+0.002, +0.061]	+0.084 [+0.027, +0.135]
occurrence contribution S_{occ}	+0.029 [+0.009, +0.050]	+0.075 [+0.043, +0.104]
intensity contribution S_{int}	+0.007 [+0.005, +0.009]	+0.013 [+0.010, +0.017]
occurrence $\times$ intensity interaction	−0.005	−0.004
occurrence BSS (unified π / standalone logistic)	+0.036 / +0.038	+0.116 / +0.117